Wind Tunnel Prototype and CFD Simulation

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A wind tunnel is an apparatus for producing a controlled stream of air for conducting aerodynamic experiments. The main goal is to simulate the behavior of the tested object when it is being hit by air while travelling at high speed. To do so, we need to create a laminar flux of air travelling at the testing speed.

1 Airflow

First of all, we need to generate a movement of air; to achieve this, we use a vacuum cleaner attached to the final end of our structure.

2 Nozzle

The entering air has to be sped up, so we place a converging nozzle up front. The fluid accelerates due to the **Continuity Equation** for incompressible flow:

$$A_{\text{entrance}} \cdot v_{\text{entrance}} = A_{\text{exit}} \cdot v_{\text{exit}} \tag{1}$$

As we can see, since the air density ρ is constant and $A_{\text{entrance}} > A_{\text{exit}}$, it follows that the exit velocity is strictly greater than the entrance velocity ($v_{\text{exit}} > v_{\text{entrance}}$).

Consequently, according to **Bernoulli's Principle**, this increase in velocity results in a decrease in static pressure:

$$P_{\text{entrance}} + \frac{1}{2}\rho v_{\text{entrance}}^2 = P_{\text{exit}} + \frac{1}{2}\rho v_{\text{exit}}^2 \tag{2}$$

3 Lamination Valve

After speeding up the fluid, we need to transform the turbulent flow into a laminar flow, in which the fluid particles follow smooth paths in layers, as shown in the following image:

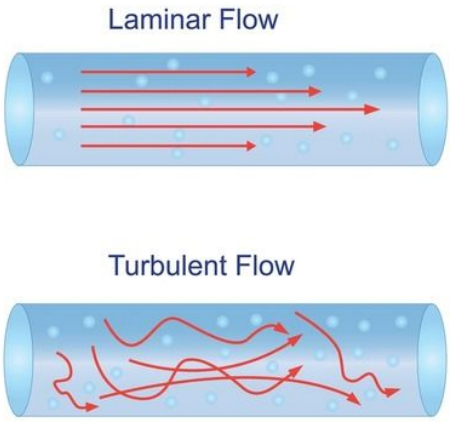


Figure 1: Laminar and turbulent flow

The component used to achieve such flow is called a Lamination Valve. In the following image, you can see the profile perpendicular to the flow of such a valve:

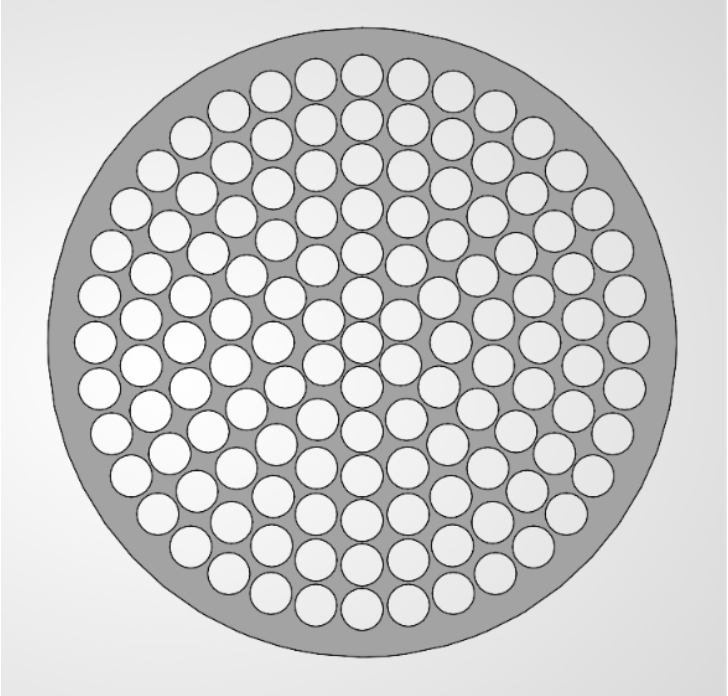


Figure 2: Lamination Valve profile

Due to 3D modelling performance issues, the percentage of area available for the fluid to pass in the CAD simulation is less than the one available in real life; therefore, the real wind tunnel will work significantly better than the simulation. After the testing chamber, we laminate the fluid once again, and at the end of the second valve, we place

another converging nozzle to attach the vacuum cleaner.

4 Wind Tunnel 3D Render and Section View

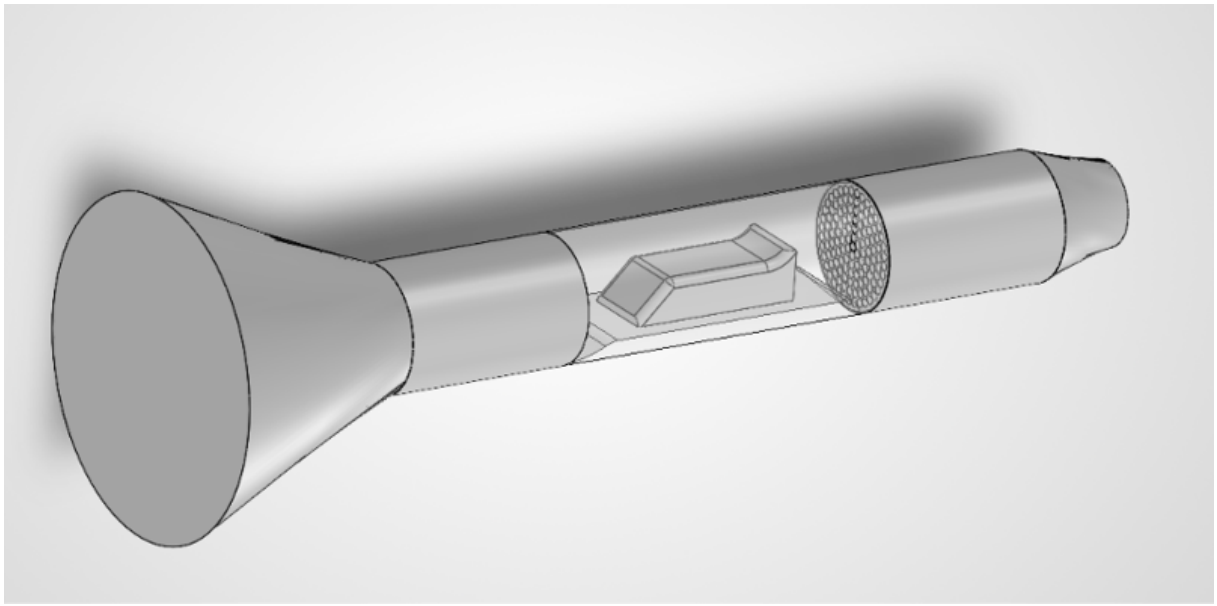


Figure 3: 3D Render

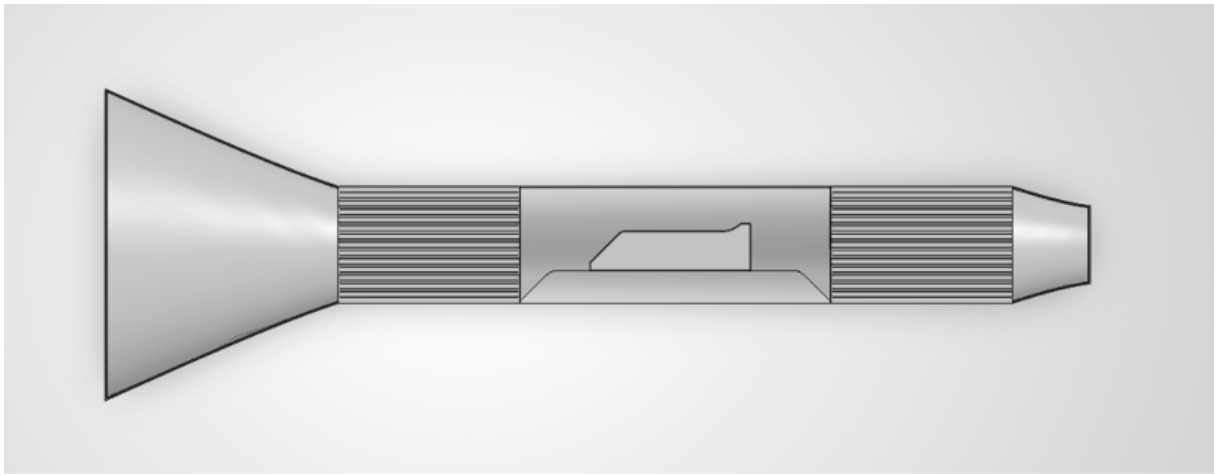
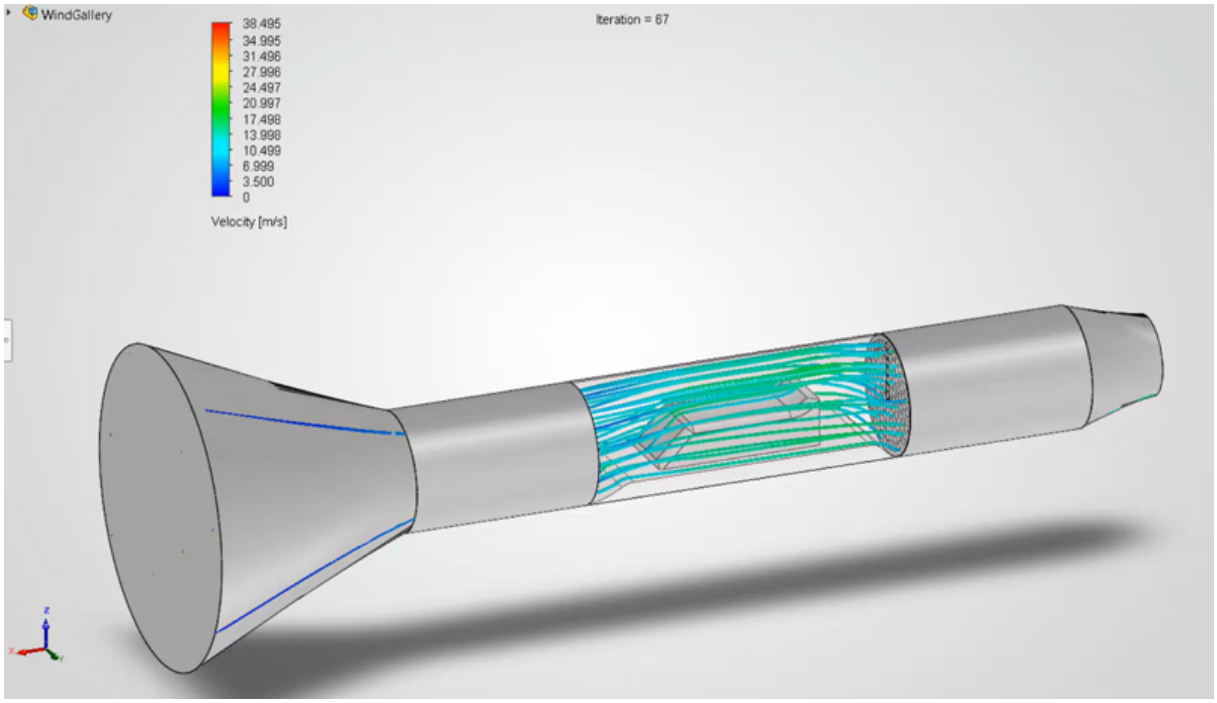


Figure 4: Section view

5 CFD Simulation

We use SolidWorks' CFD tool to simulate the wind tunnel in action. You can find the video of the model being tested in the project folder. Here is a screenshot of the simulation:



The mock model of the car occupies a significant percentage of the testing chamber, so the air is sped up on the sides of the car due to the blockage effect. This effect is acknowledged but does not compromise the structural design phase of the actual wind tunnel.